RIFAT SARWAR

Backend Engineer | Machine Learning | CSE Graduate @ IUT

PROFILE

I try to take my work seriously, but not myself. Love to code. Fueled by curiosity and occasional panic.

CONTACT

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EDUCATION

BSc in Computer Science & Engineering
Islamic University of Technology (IUT)
2022 – 2026
CGPA: 3.78 / 4.00

SKILLS

Languages
Python, Java, Go, TypeScript, C++, Dart, SQL

Backend
FastAPI, Spring Boot, Node.js / NestJS, REST, gRPC, WebSockets, microservices

Databases & Search
PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Messaging & Async
Kafka, RabbitMQ, Redis Pub/Sub, AWS SQS / SNS, Celery

AWS
EC2, ECS, EKS, ECR, Lambda, API Gateway, S3, RDS, CloudFront, IAM, VPC, CloudWatch

DevOps
Docker, Kubernetes, Helm, Terraform, GitHub Actions CI/CD, Nginx, Prometheus, Grafana, DigitalOcean, Vercel

Machine Learning
PyTorch, Hugging Face Transformers, PEFT / LoRA, vLLM, llama.cpp, quantization, distillation, RAG, LLM agents

Vector Search
pgvector, Qdrant, FAISS

Mobile & Frontend
Flutter, Next.js, React

WORK EXPERIENCE

Star Technologies — Mar 2025 – Present

Senior Software Engineer

- Own backend architecture end to end, from system design and API contracts to deployment and monitoring in production.
- Design and build **microservice backends** in **FastAPI, Spring Boot and NestJS**, deployed on **AWS (ECS / EKS)** with **Docker, Kubernetes** and **Terraform**-managed infrastructure.
- Built **real-time chat and notification services** on **WebSockets**, with **Redis** for caching and pub/sub and **Kafka / RabbitMQ** for event-driven communication between services.
- Run asynchronous workloads (notifications, data-analysis jobs) on **background workers and queues**.
- Set up **CI/CD with GitHub Actions** and observability with **Prometheus, Grafana and CloudWatch**.
- Shipped **ML services to production**: a **RAG pipeline** over vector search and a **voice assistant**, both exposed as backend APIs.
- **Key projects:**
  - **GenieA** – with *Sylhet Metropolitan Police, BD*. One backend serving a citizen app, a police app and admin panels for a **10K+ user base**. Play Store | App Store
  - **Health Star** – all-in-one healthcare app. Built the MVP from scratch (backend, mobile app and admin panel); now under active development. Play Store | App Store

Star Watch (under Star Technologies) — Nov 2025 – Present

Project Manager / Developer

- Building **drone systems from scratch** with **MAVLink, ROS, ArduPilot** and **WebRTC** video streaming.
- Automation pipeline using **LiDAR, optical flow and sonar**, with **AI-based mission planning**.

Megamind Tech (Remote – Canada) — Feb 2024 – Jul 2024

LLM Developer / ML Engineer

- **Fine-tuned LLMs** for task-specific applications with **Hugging Face Transformers** and **PEFT / LoRA**.
- Optimized **offline inference** with **vLLM** and **llama.cpp**, using **4/8-bit quantization** to fit models on resource-constrained hardware.
- Served models behind **REST inference APIs** for downstream systems.

RESEARCH

The Presence Bottleneck: Domain Adaptation and Low-End Deployment of SAM 3

BSc Thesis, IUT • Supervisor: Dr. Md. Hasanul Kabir — 2026

- Re-evaluated **SAM 3** domain adaptation and efficiency methods under a **pre-registered protocol** with bootstrap CIs and matched controls.

Robotics
ROS, MAVLink, ArduPilot, WebRTC, MuJoCo

ACHIEVEMENTS

6th Globally
International Rover Challenge (IRC)
1st in Bangladesh • 1st in Science Task

16th Globally
European Rover Challenge (ERC)

Competed
University Rover Challenge (URC)

2nd Place
CodeRush Hackathon

- A data-free prompt-embedding rebalancing raised **underwater segmentation by +0.090 mIoU**. **INT8 quantization** runs SAM 3 in 657 MB / 2.5 GB RAM with no measurable accuracy loss.
- Output-level **knowledge distillation** raised a small backbone from **0.339 to 0.486 mIoU** at the same size and speed.

From Prompt to Kinematics: Self-Verifying Generation of Valid, Controllable Robot Models

Unpublished manuscript **2026**

- **Agentic LLM system** that writes **MuJoCo (MJCF)** robot models from text or reference images. After every edit, a verifier compiles the model and step-probes it for instability.
- **pass@1 = 0.92** with zero invalid models on a 25-prompt benchmark, compared with 0.32 for direct LLM generation. Includes a vision-based judge that scores geometry and pose.

Certified Records: An Exact Audit of AI-Claimed Mathematical Constructions

Unpublished manuscript **2026**

- Re-verified **115 "new best-known" record claims** from 10 AI discovery systems (incl. **AlphaEvolve**) in **exact rational and interval arithmetic**.
- Found printed values that do not hold, records that do not beat earlier public constructions, and a **leaderboard verifier exploit** (true score 0.214 vs. 0.563 displayed).
- Computer-assisted **existence proofs for 61 spherical t-designs** (Krawczyk interval-Newton).

Rotation-Robust Bangla Sign Language Recognition from a Wrist-Referenced 6-IMU Glove

Unpublished manuscript **2026**

- Built an **ESP32 glove with six MPU6050 IMUs**. Measuring finger orientation relative to the wrist makes the features **exactly invariant** to global hand rotation.
- **Zero macro-F1 drop** under an SO(3) rotation stress test across RF, SVM and a CNN–BiLSTM–attention model. Absolute-orientation baselines lost 0.82–0.87.

TEAM PROJECTS

Project Aero **2024 – 2025**

Team Lead

- **Flight control**, **navigation** and **3D mapping** of environments from drone footage.
- CV pipelines for **real-time object detection and tracking**.

Project Altair (IUT Mars Rover Team) **2023 – 2024**

Software Sub-team Senior Developer

- Competed internationally at **IRC** (6th), **ERC** (16th) and **URC**.
- **Four-wheel drive** and **6-joint arm** control with **inverse kinematics**.
- **Object recognition** on a **custom dataset** to drive grasping, and **autonomous navigation** with obstacle avoidance.